

EMAIL THREAT INVESTIGATION ANALYSIS

Prepared by: ADENIYI MICHAEL OLAOLUWA

Institution: University of Ibadan Distance Learning Centre

SIWES Placement / Organization: Skillboost Africa

Date of Submission: 15/08/2026

EXECUTIVE SUMMARY

This investigation reviewed 13 emails received by the recipient to determine whether each was safe, suspicious, or malicious. Each email was analysed against its authentication headers (SPF/DKIM/DMARC), sender infrastructure, message content, and any embedded URLs or attachments. Of the 13 emails, 5 were assessed as SAFE, 6 as SUSPICIOUS, and 2 as MALICIOUS. The suspicious and malicious emails fall into four threat categories: credential-harvesting phishing, Business Email Compromise (BEC), an advance-fee scam, and macro-based malware delivery. The dominant pattern across the flagged emails is social engineering — urgency, secrecy, authority impersonation, and requested action — used to manipulate the recipient into clicking a link, opening an attachment, or moving money without independent verification.

EMAIL 1: PASSWORD EXPIRY REMINDER

Field	Value
From:	IT Helpdesk – Solvex Industries <ithelpdesk@solvexindustries.com>
Reply To:	ithelpdesk@solvexindustries.com
Return-path:	ithelpdesk@solvexindustries.com
Subject:	Password Expiry Reminder – Action Needed Within 5 Days
SPF:	Pass
DKIM:	Pass
DMARC:	Pass
Encryption:	Yes (TLS 1.3)
Sender IP:	mail.solvexindustries.com (192.168.14.22)

Header Red Flags

- No Suspicious activities

Content Analysis

- Observing the last paragraph “If you face any difficulty resetting your password, please raise a ticket through the IT Service Portal or contact the Helpdesk extension 204 during working hours. Do not share your password with anyone, including IT staff, over email, call or chat.”, it states categorically that the receiver should not in any form share his/her passwords. This shows the mail is not suspicious or malicious.

URLs/Attachment

- No URLs or File attachment in the mail

Verdict

- The mail is **SAFE**.

Key IOC(s)

- No Indicator of Compromise in the mail

EMAIL 2: URGENT: YOUR MAILBOX STORAGE IS FULL

Field	Value
From:	IT Support Desk <support@solvex-industries-helpdesk.com>
Reply To:	recovery-team@mail-secure-verify.net
Return-path:	bounce@mail-secure-verify.net
Subject:	URGENT: Your Mailbox Storage is FULL – Verify Now to Avoid Suspension
SPF:	Fail
DKIM:	Fail
DMARC:	Fail (Reject Policy Overridden)
Encryption:	No (Unencrypted)
Sender IP:	unknown-host-193.41.77.108.static-cloud.ru (193.41.77.108)

Header Red Flags

- Reply To mismatch with the Reply-path
- SPF/DKIM/DMARC all failed
- No Encryption which affects the integrity of the mail
- Unknown Sender IP

Content Analysis

- Urgency and Requested action in the second paragraph “To avoid suspension of your account, you are required to verify your identity IMMEDIATELY by clicking the secure link below and confirming your login credentials. This process takes less than 2 minutes.”, the word IMMEDIATELY shows urgency and the phrase “by clicking the secure link” shows requested action.
- Threat is clearly visible in the third paragraph in the sentence “Failure to verify within 24 hours will result in permanent loss of access to your mailbox and all stored data”- a fear-based consequence to ensure compliance.

URLs/Attachment

- This URL “hxxp://solvex-industries-helpdesk.verify-account-secure.com/mailbox/confirm?user=aditya.rao” was attached.

Verdict

- The mail is **SUSPICIOUS**
- It is phishing mail manipulating the recipient to click on the link to input password.

Key IOC(s)

- The attached URL
- Social Engineering (Urgency, Requested actions and Threat)

EMAIL 3: INVOICE SI-4471 & UPDATED BANK DETAILS

Field	Value
-------	-------

From:	Ramesh Kulkarni – Accounts, Apex Steel Traders <ramesh.kulkarni@apex-steeltraders.co>
Reply To:	accounts.payments@apexsteel-billing.com
Return-path:	ramesh.kulkarni@apex-steeltraders.co
Subject:	Invoice SI-4471 & Updated Bank Details for Future Payments
SPF:	Pass (for apex-steeltraders.co)
DKIM:	Pass
DMARC:	Pass
Encryption:	Yes (TLS 1.2)
Sender IP:	mail.apex-steeltraders.co (103.22.8.190)

Header Red Flags

- Reply To mismatch with the Reply-path

Content Analysis

- The second paragraph shows payment diversion in “Kindly note that all payments from this point forward should be made to our NEW account, details of which are mentioned in the attached invoice”. Changing bank details by email, combined with the Reply-To mismatch, is the classic pattern of a vendor-invoice / payment-diversion Business Email Compromise (BEC), even though the sending domain itself passed authentication.

URLs/Attachment

- The invoice was attached in pdf format “Invoice_SI-4471_Apex_Steel.pdf”

Verdict

- The mail is **SUSPICIOUS**, the recipient should independently verify any change of bank details with Apex Steel Traders through a known phone number or prior contact, not by replying to this email.

Key IOC(s)

- Reply To mismatch

EMAIL 4: REVISED LEAVE & WORK-FROM-HOME POLICY

Field	Value
From:	Human Resources – Solvex Industries <hr@solvexindustries.com>
Reply To:	hr@solvexindustries.com
Return-path:	hr@solvexindustries.com
Subject:	Revised Leave & Work-From-Home Policy – Effective 15 July 2026
SPF:	Pass
DKIM:	Pass
DMARC:	Pass
Encryption:	Yes (TLS 1.3)
Sender IP:	mail.solvexindustries.com (192.168.14.19)

Header Red Flags

- No Suspicious activities

Content Analysis

- The content of the mail only gives instructions and steps to be taken no form of social engineering is found in the mail.

URLs/Attachment

- No URLs or Attachments

Verdict

- The mail is **SAFE**.

Key IOC(s)

- No IOC(s)

EMAIL 5: CONFIDENTIAL – IMMEDIATE WIRE TRANSFER REQUIRED

Field	Value
From:	Rajeev Malhotra (Managing Director) <rajeev.malhotra.md@gmail-corpmail.com>
Reply To:	rajeev.malhotra.md@gmail-corpmail.com
Return-path:	rajeev.malhotra.md@gmail-corpmail.com
Subject:	Confidential – Immediate Wire Transfer Required (Time Sensitive)
SPF:	None (domain has no SPF record)
DKIM:	None
DMARC:	None
Encryption:	No (Unencrypted)
Sender IP:	smtp-relay-88.freemailhost.io (45.137.22.9, Netherlands)

Header Red Flags

- SPF/DKIM/DMARC are not found
- No Encryption which affects the integrity of the mail
- Mail subject line includes pressure words like “Confidential” and “Time Sensitive”

Content Analysis

- The content of mail includes Urgency and Secrecy in the statement “Do not discuss this with Priya or anyone in Accounts for now — I want to keep this quiet until the deal is signed.”
- It also shows Impersonation, the sender is impersonating the managing director of the company

URLs/Attachment

- No URLs or Attachments

Verdict

- The mail is **SUSPICIOUS**
- It is a phishing mail manipulating the recipient to send urgent money to an undisclosed account
- The recipient should not acknowledge the mail so the undisclosed account won't be sent.

Key IOC(s)

- The Mail Subject line pressure
- Social Engineering (Urgency, Requested actions and Impersonation)

EMAIL 6: 'Q1_Vendor_Compliance_Report.xlsx' HAS BEEN SHARED WITH YOU

Field	Value
From:	Google Drive <drive-shares-noreply@google.com>
Reply To:	drive-shares-noreply@google.com
Return-path:	drive-shares-noreply@google.com
Subject:	'Q1_Vendor_Compliance_Report.xlsx' has been shared with you
SPF:	Pass
DKIM:	Pass
DMARC:	Pass
Encryption:	Yes (TLS 1.3)
Sender IP:	mail-sor-f89.google.com (209.85.220.89)

Header Red Flags

- No Suspicious Activities

Content Analysis

The mail is a standard Google Drive share notification, with a note referencing a prior conversation with the sharer. There is no urgency, no credential request, and headers fully authenticate against Google's own infrastructure

URLs/Attachment

- The URL “https://docs.google.com/spreadsheets/d/1kQ3z-example-fileid-9f2/edit?usp=sharing” was attached for easy access to the shared documents.

Verdict

- The mail is **SAFE**
- In as much there is a previous conversation with the said person the recipient can click on the link.

Key IOC(s)

- No IOC(s)

EMAIL 7: UNUSUAL SIGN-IN ACTIVITY DETECTED.

Field	Value
From:	Microsoft account team <security-noreply@micros0ft-online.com>
Reply To:	no-reply@micros0ft-online.com
Return-path:	bounce@micros0ft-online.com
Subject:	Unusual sign-in activity detected on your Microsoft account
SPF:	Fail
DKIM:	Fail
DMARC:	Fail
Encryption:	No
Sender IP:	host-77-91-134-6.vpn-exit.net (77.91.134.6, unknown origin)

Header Red Flags

- Reply To mismatch with the Reply-path
- SPF/DKIM/DMARC all failed
- No Encryption which affects the integrity of the mail
- Unknown Sender IP
- Domain typosquat: “micros0ft-online.com” uses a zero in place of the letter ‘o’ in Microsoft.

Content Analysis

- Urgency and Requested action in the second and third paragraph which indicates social engineering in a well-orchestrated manner with them knowing fully that no one has sign-in into his Microsoft account, particularly the third paragraph which will prompt the recipient to click the link, “If you don't recognise this activity, we strongly recommend you review your account immediately to prevent unauthorized access to your email, OneDrive files and Teams messages.”, the Urgency and Requested action is embedded in the statement.

URLs/Attachment

- This URL “hxxp://login.micros0ft-online.com/secure/verify-identity?ref=aditya.rao” was attached.

Verdict

- The mail is **SUSPICIOUS**
- It is phishing mail impersonating Microsoft to harvest login details.

Key IOC(s)

- The attached URL (Microsoft with ‘0’ instead of ‘o’)
- Social Engineering (Impersonation, Urgency and Requested actions)
- Reply To mismatch

EMAIL 8: EXCITING CAREER OPPORTUNITY

Field	Value
From:	Neha Kapoor, HR Consultant <neha.kapoor.hr@talentbridge-recruiters.net>
Reply To:	neha.kapoor.hr@talentbridge-recruiters.net
Return-path:	neha.kapoor.hr@talentbridge-recruiters.net
Subject:	Exciting Career Opportunity – Senior Finance Role (Immediate Response Needed)
SPF:	Pass
DKIM:	Pass
DMARC:	None (No Policy Published)
Encryption:	Yes (TLS 1.2)
Sender IP:	mail.talentbridge-recruiters.net (156.213.44.71, Singapore)

Header Red Flags

- DMARC cannot be found
- Mail subject line contains a statement of urgency (Immediate Response Needed)

Content Analysis

- The mail content show requested action asking the recipient to download an attached file and send some of his personal information in the statement “long with a scanned copy of your latest salary slip and PAN card for our internal verification process”, a career opportunity asking for latest payment slip and PAN card is suspicious.

URLs/Attachment

- An attached zip file “JD_SeniorFinanceManager_Client.zip”

Verdict

- The mail is **MALICIOUS**, a job description file sized up to 2.4mb as a zip file definitely contains more than the job description but this can’t be justified until malware analysis is performed.
- It is phishing mail manipulating the recipient to download the attached file and send latest payment slip and PAN card.

Key IOC(s)

- The attached zipped file
- Mail subject line which is showing urgency

EMAIL 9: YOUR TALLYPRIME CLOUD SUBSCRIPTION RENEWAL INVOICE

Field	Value
From:	Tally Solutions Billing <billing@tallysolutions.com>
Reply To:	billing@tallysolutions.com
Return-path:	billing@tallysolutions.com
Subject:	Your TallyPrime Cloud Subscription Renewal Invoice – INV-88213

SPF:	Pass
DKIM:	Pass
DMARC:	Pass
Encryption:	Yes (TLS 1.3)
Sender IP:	mta-out.tallysolutions.com (203.0.113.44)

Header Red Flags

- No Suspicious activities

Content Analysis

- Paragraph two “Payments can be made through your existing registered payment method on the Tally Customer Portal, or via NEFT/RTGS to the bank account already on file with our billing team — no bank details have changed.” shows the realness of mail, no suspicious activities
-

URLs/Attachment

- Attached pdf file “INV-88213_TallyPrime_Renewal.pdf” with size 146kb

Verdict

- The mail is **SAFE**.

Key IOC(s)

- No Indicator of Compromise in the mail

EMAIL 10: CONGRATULATIONS!!! YOU HAVE WON USD 1,000,000 IN THE INTERNATIONAL EMAIL LOTTERY

Field	Value
From:	INTERNATIONAL LOTTERY BOARD <claims.department009@yandex-mailer.com>
Reply To:	agent.williams.claims@outlook-verify.info
Return-path:	claims.department009@yandex-mailer.com
Subject:	CONGRATULATIONS!!! YOU HAVE WON USD 1,000,000 IN THE INTERNATIONAL EMAIL LOTTERY
SPF:	Fail
DKIM:	Fail
DMARC:	Fail
Encryption:	No
Sender IP:	unknown-relay-node4.freehosting-mail.com (185.220.101.44, Netherlands)

Header Red Flags

- SPF/DKIM/DMARC all failed
- No encryption

- Reply To and Return-path mismatch
- Unknown sender IP

Content Analysis

- “A small processing and courier fee of USD 250 will be required to release your winning certificate and cheque”, this sentence shows social engineering manipulating the recipient to pay a sum USD 250 for a non-existential amount of USD 1,000,000.

URLs/Attachment

- An attached URL “hxxp://claim-your-prize-now.lottery-verify.info/claim?ref=44921”

Verdict

- The mail is **SUSPICIOUS**
- The recipient should not click on the embedded link.

Key IOC(s)

- Reply To mismatch
- Unknown sender IP

EMAIL 11: QUICK REQUEST BEFORE MY FLIGHT

Field	Value
From:	Suresh Iyer (CFO) <suresh.iyer@solvexindustries.com>
Reply To:	s.iyer.cfo.travel@outlook.com
Return-path:	suresh.iyer@solvexindustries.com
Subject:	Quick request before my flight
SPF:	Pass (envelope sender matches solvexindustries.com)
DKIM:	Pass
DMARC:	Pass
Encryption:	Yes (TLS 1.3)
Sender IP:	mail.solvexindustries.com (192.168.14.19) — header shows internal relay, but Reply-To routes externally

Header Red Flags

- Reply To mismatch with the Reply-path
- Sender IP Header show Reply To routes externally

Content Analysis

- Urgency was clearly shown in the last sentence in the first paragraph “Need it in the next 20 minutes before boarding closes”
- The last paragraph show secrecy telling the recipient not to call the office line but speak to him through mail.

URLs/Attachment

- No URLs/Attachment found

Verdict

- The mail is **SUSPICIOUS**
- This is a CFO-fraud Business Email Compromise (BEC) attempting to pressure the recipient into purchasing five Amazon gift cards worth Rs. 5,000 each. Even though SPF/DKIM/DMARC pass on the visible from address, the externally-routed Reply-To is the key giveaway.

Key IOC(s)

- Reply To mismatch
- Social Engineering (Urgency and Secrecy)

EMAIL 12: REMINDER: MANDATORY SECURITY AWARENESS TRAINING

Field	Value
From:	Information Security Team – Solvex Industries <infosec@solvexindustries.com>
Reply To:	infosec@solvexindustries.com
Return-path:	infosec@solvexindustries.com
Subject:	Reminder: Mandatory Security Awareness Training – Due This Friday
SPF:	Pass
DKIM:	Pass
DMARC:	Pass
Encryption:	Yes (TLS 1.3)
Sender IP:	mail.solvexindustries.com (192.168.14.19)

Header Red Flags

- No Suspicious activities

Content Analysis

- “To access the module, log in to learn.solvexindustries.com using your employee ID. If you experience any login issues, please raise a request with the IT Helpdesk through the official portal.”, this sentence shows clearly that there are no suspicious activities in the mail.

URLs/Attachment

- No URLs or Attachment

Verdict

- The mail is **SAFE**.

Key IOC(s)

- No Indicator of Compromise in the mail

EMAIL 13: PENDING INVOICE – PLEASE REVIEW & PROCESS ON PRIORITY.

Field	Value
From:	Accounts Receivable <accounts.receivable@shreeganesh-logistics.in>
Reply To:	accounts.receivable@shreeganesh-logistics.in
Return-path:	mailer@shreeganesh-logistics.in
Subject:	Pending Invoice – Please Review & Process on Priority
SPF:	Fail (domain not authorised to send)
DKIM:	Fail
DMARC:	Fail
Encryption:	No
Sender IP:	webmail-host-4.sharedhosting-cluster.com (41.203.88.17, unrelated hosting cluster)

Header Red Flags

- Reply To mismatch with the Reply-path
- SPF/DKIM/DMARC all failed
- No Encryption which affects the integrity of the mail
- SPF shows clearly that the domain is not authorized to send

Content Analysis

- Second paragraph, the last sentence shows Requested action while claiming that an older billing system was used “Please enable editing/macros if prompted so all the invoice tables display correctly, as the file was generated on an older billing system.”. Asking a recipient to enable macros is a well-known technique used to trigger malicious VBA code on open.

URLs/Attachment

- This URL “hxxp://shreeganesh-logistics.in.invoice-view-secure.com/download?file=inv2207” was attached.
- A word document (macro-enabled) “Pending_Invoice_Challan_Details.docm” of size 1.1mb was attached

Verdict

- The mail is **MALICIOUS**
- The URL shows a link to download a file that was not stated in the mail.
- The attach word document is also macro-enabled while requesting the recipient to enable editing/macros if prompted shows clearly the file is embedded with malware.

Key IOC(s)

- The attached URL
- The attached word documents
- Social Engineering (Requested actions)
- Reply To mismatch

FINAL SUMMARY

No	Sender	Verdict	Threat Type	Key IOC
Email 1	IT Helpdesk – Solvex Industries <ithelpdesk@solvexindustries.com>	SAFE	—	None
Email 2	IT Support Desk <support@solvex-industries-helpdesk.com>	SUSPICIOUS	Phishing	Attached URL
Email 3	Ramesh Kulkarni – Accounts, Apex Steel Traders <ramesh.kulkarni@apex-steeltraders.co>	SUSPICIOUS	BEC (Payment Diversion)	Reply To mismatch
Email 4	Human Resources – Solvex Industries <hr@solvexindustries.com>	SAFE	—	None
Email 5	Rajeev Malhotra (Managing Director) <rajeev.malhotra.md@gmail-corpmail.com>	SUSPICIOUS	BEC	Social Engineering
Email 6	Google Drive <drive-shares-noreply@google.com>	SAFE	—	None
Email 7	Microsoft account team <security-noreply@microsoft-online.com>	SUSPICIOUS	Phishing	Attached URL
Email 8	Neha Kapoor, HR Consultant <neha.kapoor.hr@talentbridge-recruiters.net>	MALICIOUS	Malware delivery	Attached File
Email 9	Tally Solutions Billing <billing@tallysolutions.com>	SAFE	—	None
Email 10	INTERNATIONAL LOTTERY BOARD <claims.department009@yandex-mailer.com>	SUSPICIOUS	Phishing	Reply To mismatch
Email 11	Suresh Iyer (CFO) <suresh.iyer@solvexindustries.com>	SUSPICIOUS	BEC	Reply To mismatch
Email 12	Information Security Team – Solvex Industries <infosec@solvexindustries.com>	SAFE	—	None
Email 13	Accounts Receivable <accounts.receivable@shreeganesh-logistics.in>	MALICIOUS	Malware delivery	Attached File

Overall Findings

- 2 Malicious Mails
- 5 Safe Mails
- 6 Suspicious Mails

Common pattern observed: the common pattern observed is basically social engineering, most of the mails are just trying to manipulate him to perform one task or the other

Recommendation: Enrol the recipient in phishing/BEC-focused security awareness training, with particular emphasis on verifying payment or wire-transfer requests through an independent channel (phone call to a known number) rather than by replying to the email itself. Block the identified malicious domains and URLs at the email gateway, and quarantine the two malicious emails and their attachments organization-wide in case they were sent to other recipients.